

1. Overview:

```
SELECT
    FACULTY_ID,
    FIRST_NAME || ' ' || LAST_NAME AS FACULTY_NAME,
    HIRE_DATE,
    EMAIL,
    PHONE_NUMBER,
    JOB_ID
FROM AD.AD_FACULTY_DETAILS
ORDER BY FACULTY_ID;
```

We see this query as necessary to view the basic data of all faculty members, such as their identification, full name, etc.

It is relevant because it allows us to see the faculty's structure.

2. Faculties and courses offered:

```
SELECT
    AD_FACULTY_DETAILS.FACULTY_ID,
    AD_FACULTY_DETAILS.FIRST_NAME || ' ' || AD_FACULTY_DETAILS.LAST_NAME AS
FACULTY_NAME,
    AD_FACULTY_DETAILS.JOB_ID,
    AD_COURSE_DETAILS.COURSE_ID,
    AD_COURSE_DETAILS.COURSE_NAME
FROM AD.AD_FACULTY_DETAILS,
    AD.AD_FACULTY_COURSE_DETAILS,
    AD.AD_COURSE_DETAILS
WHERE AD_FACULTY_DETAILS.FACULTY_ID = AD_FACULTY_COURSE_DETAILS.FACULTY_ID
    AND AD_FACULTY_COURSE_DETAILS.COURSE_ID = AD_COURSE_DETAILS.COURSE_ID
ORDER BY AD_FACULTY_DETAILS.FACULTY_ID, AD_COURSE_DETAILS.COURSE_ID;
```

With this query, we aim to link each professor to the specific courses they teach. To do this, we cross-referenced the faculty tables, course assignments, and course details. We found this relevant because it clarifies and answers our question about the crucial courses each professor teaches.

3. Number of courses per faculty:

```
SELECT
    AD_FACULTY_DETAILS.FACULTY_ID,
    AD_FACULTY_DETAILS.FIRST_NAME || ' ' || AD_FACULTY_DETAILS.LAST_NAME AS
FACULTY_NAME,
    AD_FACULTY_DETAILS.JOB_ID,
    COUNT(AD_FACULTY_COURSE_DETAILS.COURSE_ID) AS COURSES_ASSIGNED
FROM AD.AD_FACULTY_DETAILS, AD.AD_FACULTY_COURSE_DETAILS
WHERE AD_FACULTY_DETAILS.FACULTY_ID = AD_FACULTY_COURSE_DETAILS.FACULTY_ID(+)
GROUP BY AD_FACULTY_DETAILS.FACULTY_ID, AD_FACULTY_DETAILS.FIRST_NAME,
    AD_FACULTY_DETAILS.LAST_NAME, AD_FACULTY_DETAILS.JOB_ID
ORDER BY COURSES_ASSIGNED DESC;
```

We count how many courses each professor has, which is important to identify the balance and equilibrium in the distribution of work, that is, whether there are professors with more courses than others.

4. Type of work:

```
SELECT
    JOB_ID,
    COUNT(FACULTY_ID) AS NUMBER_OF_FACULTIES,
    MIN(HIRE_DATE) AS EARLIEST_HIRE,
    MAX(HIRE_DATE) AS LATEST_HIRE
FROM AD.AD_FACULTY_DETAILS
GROUP BY JOB_ID
ORDER BY NUMBER_OF_FACULTIES DESC;
```

5. Seniority of faculty staff:

```
SELECT
    JOB_ID,
    COUNT(FACULTY_ID) AS TOTAL_FACULTIES,
    ROUND(AVG(SYSDATE - HIRE_DATE), 0) AS AVG_TENURE_DAYS,
    ROUND(AVG(SYSDATE - HIRE_DATE) / 365, 1) AS AVG_TENURE_YEARS,
    MIN(HIRE_DATE) AS MOST_SENIOR,
    MAX(HIRE_DATE) AS MOST_RECENT
FROM AD.AD_FACULTY_DETAILS
GROUP BY JOB_ID
ORDER BY AVG_TENURE_DAYS DESC;
```

We would like to know the average seniority of the staff; this could help us in different ways, such as considering promotions, identifying staff nearing retirement (finding replacements), or determining, if there is too much new staff, what might have caused the departure of the previous staff.